

The Clock Is in the Frame, the Wall Is Not

Read-Frame Covariance, Affine Invariants, and the Predictable Instrument
Wall in the Christoffel Rank Read

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Abstract

We report an operational sorting of the Christoffel Rank Read into read-frame quantities and object quantities, obtained by measurement rather than stipulation. Three preregistered experiment sets were run against the Hankel/LDL^T instrument. First, a clock law: the precision cost of the factorization is $c = 2 \cdot \log_{10}(r_{\text{out}}/\text{cap})$ digits per resolved pivot — a conformal invariant of the support taken jointly with the read origin — verified within 10% on four node families, with a shared -5% systematic identified as the finite-index Szegő correction. Second, the instrument wall of Part II-B is shown to be a predictable function, $W_{\text{instr}}(\text{dps}) = D^{-1}(\text{dps} - \kappa) + G$ with $\kappa = 0 \pm 1$: calibrated on one family, it predicted false-wall locations blind on two others to within the preregistered ± 2 band, including across a regime break, and in all six starved runs the wall was announced by the first negative sub-floor pivot — a sign flip — re-deriving the detector law $W = k_{\text{onset}} + G$ from the depth side. Third, an affine sort: under translation, scaling, and reflection of the read-frame, the object wall is exactly invariant, the Jacobi α 's shift by exactly the frame, the β 's are frame-blind, and the clock covaries by the stated law. The sort refutes, inside this system, the strong relational thesis that no coordinate-free quantities exist, while confirming that every continuous instrument-side quantity is a ratio, a difference, or a sign. What survives every admissible frame is an integer — the count of distinctions — and the paper closes with the finite-to-infinite passage of that invariant, where a count must become a counting function, as the gated Clay-adjacent experiment.

1. Anchor and scope

Of the Clay problems, this instrument touches two. On the BSD side, both channels — rank certificates and leading coefficients — already reproduce from a_p alone (Scale Notebook, Cells 1–2). On the RH side, the pending experiment feeds prime-derived moments at growing budgets and asks whether the wall tracks the zero-counting function $N(T)$. That run is unreadable until the instrument wall can be separated from the object wall at any finite budget. This paper builds that separation, and the separation forces something larger: a sorting of every quantity in the program into read-frame and object.

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github.com/QuHarmonics/The-Nexus-Harmonic-Reality info@quharmonics.com

Part II-B established that the wall is three walls — object (algebraic), instrument (precision floor), detector (stochastic sign statistics). This paper makes the instrument wall quantitative (§4), identifies the detector wall's announcement mechanism (§4), gives the instrument a clock law (§§2–3), and adds a new axis: the read-frame (§5). The synthesis (§6) sorts the results into a three-layer architecture — representation, invariant, execution — whose articulation arose in the cross-model dialogue of this session; the measurements are what earn it a place in the record.

2. The clock law

Let $\mu_k = \sum w_i x_i^k$ be the moments of an M -point positive measure read from an origin at zero, H the $(M+1) \times (M+1)$ Hankel matrix $[\mu_{i+j}]$, and s_i the i -th pivot of its Cholesky sweep — classically $s_i = h_i$, the squared norm of the i -th monic orthogonal polynomial. Define the cancellation depth $D(i) = \log_{10}(\text{big}_i / |s_i|)$, where big_i is the largest mass entering the pivot's computation (the maximum of μ_{2i} and the subtracted sum). $D(i)$ counts the digits the instrument must destroy to resolve pivot i . Precision is therefore a tick budget: each resolved pivot is one tick, and the local slope $c = dD/di$ is the burn rate — the clock speed of the object as read from this frame.

Potential theory fixes the rate. For supports with outer radius $r_{\text{out}} = \max|x|$ (from the read origin) and logarithmic capacity cap (for an interval $[a, b]$, $\text{cap} = (b-a)/4$), Szegő asymptotics give $\mu_{2i} \sim r_{\text{out}}^{2i}$ and $h_i \sim \text{cap}^{2i}$, hence the capacity clock law: $c = 2 \cdot \log_{10}(r_{\text{out}}/\text{cap})$ digits per tick. Preregistered: P₁, each family's early-window slope within 10% of prediction; P₄, no fitted slope near 0.35 — the framework's own canon places $H = \pi/9$ in recursive compression with feedback, not enumeration geometry, so its absence here is a domain confirmation, not a strike. All runs: $M = 30$, $\text{dps} = 400$, early window $i = 3..12$, late window $i = 18..27$.

family	wall	pred c	early c	late c	break	P ₁ (10%)
random(−10,10)	31	0.626	0.636	1.367	YES	PASS
integer 1..M	31	1.234	1.191	1.605	YES	PASS
chebyshev[−1,1]	31	0.602	0.571	0.592	no	PASS
equispaced[−1,1]	31	0.602	0.571	0.979	YES	PASS

Four for four, with a −5% systematic shared by every family — the finite-index correction to the Szegő limit, a subleading term, not a challenge to the law. P₄: the eight fitted slopes are 0.636, 1.367, 1.191, 1.605, 0.571, 0.592, 0.571, 0.979; nothing lands in (0.33, 0.37). Absent, as preregistered.

One structural fact deserves its own sentence: scale cancels. Any interval centered on the read origin burns $2 \cdot \log_{10} 2 \approx 0.602$ digits per tick whether it spans $[-1, 1]$ or $[-10^{10}, 10^{10}]$. The clock reads the shape of the support and its position relative to the read-head; it cannot see size. Shape before value, as arithmetic.

3. The one-clock measure

The Chebyshev family is not merely the best-behaved row of the table; it is a theorem. Unit weights on Chebyshev nodes are proportional to the Gauss–Chebyshev quadrature weights, which are all equal (π/M). Gauss quadrature on M nodes is exact through degree $2M-1$, so the discrete moments equal (M/π) times the continuum arcsine moments exactly through μ_{2M-1} — and therefore the discrete pivots are the continuum pivots all the way to the wall. Measured: slope 0.586 over $i = 3..29$ with maximum deviation from a straight line of 0.106 digits. The equilibrium measure of the interval is the one-clock measure: it never shifts gears because there is no discreteness constraint left to bind.

Every family that deviates from the equilibrium density pays a growing surcharge — the constrained-equilibrium (Runge-side) dive. Random-uniform, integer, and equispaced families all accelerate; the integer family's local slope climbs from 1.19 to 2.24 across the sweep, and a 25%-departure threshold lands at step 21 ($t = i/M = 0.70$). Stated honestly: the acceleration is smooth, so the break step is threshold-dependent; the finding is the acceleration plus the Chebyshev exception, not a kink. Ledger hook: if the two-regime transition already recorded for the deterministic integer-node family in the Part II-B ledger sits in this territory, the constrained-equilibrium mechanism attaches to a verified anomaly; if it sits elsewhere, these are two phenomena and the record should say so. Status: the exactness argument is classical and stands as mathematics; the gear-change mechanism claim is PULL.

4. The instrument wall is a function

The claim: $W_{\text{instr}}(\text{dps}) = D^{-1}(\text{dps} - \kappa) + G$ — the false wall appears at the step where the object-side depth profile crosses the budget, less a small headroom κ , plus the detector's stochastic delay G . Protocol, fixed before running: κ is calibrated on the random family only; integer and equispaced are blind transfers; PASS is ± 2 steps.

Calibration returned $\kappa = 1.05, -0.14, 0.14$ across three starved runs: $\kappa = 0 \pm 1$. The three-sample mean prints as 0.35; the scatter is larger than the value, and the coincidence with H is recorded here precisely so that no one resurrects it. Killed on sight.

blind transfer	dps	predicted	measured	trip	verdict
integer 1..M	20	18	18	sign	PASS
integer 1..M	30	25	23	sign	PASS
equispaced[-1,1]	15	25	24	sign	PASS

Three for three, deltas 0, -2, -1 — and the second transfer crossed the integer family's regime break, which is the point: the prediction uses the measured object profile, not a linear extrapolation. The mechanism finding outranks the headline: all six starved trips — calibration and transfer — fired the sign detector; zero fired the ratio floor. The instrument wall is announced by a bit flip, the first negative pivot in the sub-floor stream. This re-derives, from the depth side, the detector law $W_{\text{observed}} = k_{\text{onset}}(\mu, \epsilon) + G$ of the v_4 statistics: the onset is deterministic (depth crossing budget), the announcement waits for the first negative

sign, and the observed deltas are G's band. Consequence: at starved budgets the ratio floor is dead weight; the detector wall of Part II-B is the sign channel.

This is also the subtraction the N(T) run needs. Measure D(i) for an arithmetic measure, predict the instrument wall, and only structure in excess of the prediction counts as object signal.

5. The affine sort

Preregistered: A1, the object wall is invariant under all five frames — translate (-15.5 , $+100$), scale ($\times 1000$), reflect ($x \rightarrow -x$); A2, the clock c is invariant under scaling and reflection (the ratio $r_{\text{out}}/\text{cap}$ cannot see them) and covariant under translation per the capacity law; A3, under translation by s every Jacobi α_k shifts by exactly s and every β_k is unchanged; under reflection $\alpha \rightarrow -\alpha$ with β fixed; under scaling by a , $\alpha \rightarrow a\alpha$ and $\beta \rightarrow a^2\beta$. Base object: integer nodes $1..30$, unit weights.

frame	wall	c meas	α_0	α_1	β_1
centered ($x-15.5$)	31	0.571	0.0	0.0	74.9167
identity (x)	31	1.191	15.5	15.5	74.9167
shift +100	31	2.461	115.5	115.5	74.9167
scale $\times 1000$	31	1.191	15500	15500	7.4917×10^7
reflect ($-x$)	31	1.191	-15.5	-15.5	74.9167

All exact — integer nodes at 400 digits leave no rounding residue, so the invariances hold to the last digit (measured deviations 0.0). A1: the object wall sat at 31 under every frame. A2: the clock is bit-identical under scaling and reflection and covariant under translation exactly as the capacity law dictates. A3: every α shifted by exactly the frame and nothing else; every β was frame-blind under translation and reflection and scaled by exactly 10^6 under the $1000\times$ dilation.

The inventory, sorted. Object-side — invariant under the affine read-group: the wall (rank; an integer, exact under all five frames); the β 's up to the scale action, so β -ratios are full affine invariants; the α -gaps up to scale, so gaps normalized by $\sqrt{\beta_1}$ are invariants. Frame-side — covariant: the absolute α 's, which carry the shift and nothing else; the clock speed; the depth profile; the false-wall location; every individual pivot. A small confirming touch: $\beta_1 = 74.9167 = (M^2-1)/12$ is the variance of the node set — the archetypal difference-quantity, translation-blind by construction.

Corollary (the geodesic frame). $c(t) = 2 \cdot \log_{10}(\max_i |x_i - t| / \text{cap})$ is minimized when the read origin sits at the Chebyshev center of the support, and for any single interval the minimum is universal: $c_{\text{min}} = 2 \cdot \log_{10} 2 \approx 0.602$ digits per tick. The excess $c - c_{\text{min}}$ is pure frame — the price of reading off-center. Verified in the data: the centered integer frame measured 0.571 against 0.602 predicted (the same -5% Szegő offset), and the far-shifted frame's excess matches $2 \cdot \log_{10}(130/14.5) = 1.905$ predicted.

The bit, re-read. A bit is not 0-or-1; it is distinguishable-versus-not. Reflection flipped the sign of every α — every label — while the wall and the β 's, the distinction structure, did not move. Labels are frame; distinctions are object. What survives every admissible frame in this system is an integer: the count of distinctions. Numbers as gap-labels, measured.

6. Three layers: representation, invariant, execution

The affine sort adjudicates two philosophies at once. Strong relationalism — there are no absolute values, only relationships — is REFUTED inside this system: the wall is coordinate-free, five frames, exact zeros. Naive absolutism fares no better: every continuous quantity on the instrument side is a ratio, a difference, a log-ratio, or a sign; none is absolute. The operational middle holds: everything observable is relative; everything real is whatever survives the relative descriptions. This is Weyl's dictum — objectivity means invariance — and Klein's Erlangen program arriving operationally: choose the group of admissible read-frames; the object is the invariants; the instrument is the covariants.

The cross-model dialogue of this session contributed a refinement the measurements support: three layers, not two. Layer 1, representation — moments, α 's, node coordinates, pivots; everything that moves under frame changes. Layer 2, invariants — the wall, the rank, the β -ratios, the α -gap structure; the mathematical object, the equivalence class across representations. Layer 3, execution — the recurrence itself: the process that changes representation while preserving the invariants, the compiler carrying the object through its expressions. The measured shadow of Layer 3 is real but modest: invariants are route-independent while cost is frame-dependent — the same object charges 0.571 digits per tick read from its center and 2.461 read from a hundred units away. Whether execution deserves ontological standing distinct from the other two layers is beyond this instrument; the layer is recorded as architecture, PULL, not as measurement.

For the record, three items from the same dialogue that do not enter the ledger. First, “nothing is stored” — corrected by Cell 4 of the Scale Notebook: storage is global, not absent; nothing is stored locally, everything is stored everywhere. Second, the mutual-measurement graph (“every object is part of every other object's frame”) — in this sandbox the read is one-way and back-action-free; the object shapes the instrument and the instrument never touches the object; the symmetric graph is a physics hypothesis with no local shadow yet. Third, light as a stream relationship — no contact points; parked at Ω .

7. The invariant at infinite support (open)

At finite support the terminal invariant is a count: wall = support + 1, exact 300 of 300 (prior work, LOCKED), affine-proof (this paper). At infinite support the count disappears, and the candidate replacement — raised in the same dialogue — is the topology of distinguishability: spectral type, essential support, counting functions in place of counts. The Clay-adjacent experiment is exactly this passage. For prime-derived moments, the question “does the wall track $N(T)$?” asks whether the count invariant becomes the zero-counting function — a count that has learned to be a function of height. The run remains gated on OP-1 (the dps of the §9.2 build-order run) and on the moment-convention agreement; what this paper contributes is the subtraction that makes it readable (§4) and the next bridge: clock speeds c for the a_p measures of the four BSD curves, asking whether c carries arithmetic content — conductor, rank — once the support is an arithmetic object.

8. Status ledger

claim	status	evidence
wall = support + 1	LOCKED (prior)	300/300 exact; re-verified in every run here
Gauss–Chebyshev exactness $\Rightarrow$ one-clock measure	LOCKED (classical)	measured residual 0.106 digits, $i = 3..29$
Jacobi frame covariance (α shifts; β blind)	LOCKED (classical)	exact zeros, five frames
capacity clock law $c = 2 \cdot \log_{10}(r_{\text{out}}/\text{cap})$	PULL	4/4 within 10%; –5% systematic identified
instrument wall $W = D^{-1}(\text{dps} - \kappa) + G$, $\kappa = 0 \pm 1$	PULL	3/3 blind transfer, ± 2 band
sign-first announcement (detector = sign channel)	PULL	6/6 starved trips; ties to v4 G
gear change = constrained-equilibrium surcharge	PULL	break $t \approx 0.70$, threshold-dependent; II-B ledger comparison pending
strong relationalism (no coordinate-free quantities)	REFUTED	A1: five frames, exact wall invariance
$\bar{\kappa} = 0.35$	KILLED	3-sample mean; scatter ± 1 exceeds value
0.35 in enumeration clock rates	ABSENT (P4)	eight slopes, none in (0.33, 0.37)
Layer-3 execution ontology	PULL (architecture)	route-independence of invariants; frame-dependence of cost

9. Reproduction

Every number above was regenerated from moments within a single session; nothing was trusted from prior output. Three scripts accompany this paper — clock_speed.py (the four-family clock and starvation runs), frame_test.py (translation covariance), affine_sort.py (the five-frame sort with Jacobi data) — mpmath 1.3.0, dps 400 for object-side profiles, every PASS/FAIL preregistered in the text that precedes it. Per program discipline, no recursive self-validation: no PULL row moves to LOCKED until the scripts are re-run independently, on separate hardware and preferably by a separate model instance.

Acknowledgment. Produced within the multi-AI cross-validation network. The three-layer articulation and the topology-of-distinguishability generalization entered through the GPT node in this round; measurement design, runs, and the sorting verdicts are the Claude node's; direction, standards, and the anchor discipline are DK's.

